Complete Tech Stack — OmniHub (MERN-Based)

Multiplatform Messaging & Automation Hub

Tech Stack Overview (One-Page View)

Layer	Technology	Language	Purpose
Frontend	React.js + Vite	JavaScript (JSX)	Dashboard UI
Styling	Tailwind CSS	CSS	UI styling
Charts	Recharts	JavaScript	Analytics graphs
State Management	Redux Toolkit / Zustand	JavaScript	Global state
Real-time Client	Socket.io Client	JavaScript	Live updates
Backend	Node.js + Express	JavaScript	API server
Real-time Server	Socket.io	JavaScript	WebSocket server
Database	MongoDB Atlas	BSON/JSON	Data storage
ODM	Mongoose	JavaScript	MongoDB modeling
AI Engine	Google Gemini API	REST/SDK	Auto-replies
Auth	JWT + bcrypt	JavaScript	Authentication
File Storage	Cloudinary / Supabase Storage	REST API	Media files
Email	Nodemailer	JavaScript	OTP, notifications
Caching	Redis (Upstash)	Key-Value	Rate limiting, sessions
Queue	BullMQ	JavaScript	Background jobs
Hosting (Backend)	Render / Railway	—	Node.js deploy
Hosting (Frontend)	Vercel	—	React deploy
Hosting (DB)	MongoDB Atlas	—	Free 512MB cluster
Tunnel	Ngrok	—	Local webhook testing

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1. FRONTEND TECH STACK

1.1 Core Framework

Tool	Version	Purpose	Why This?
React.js	18.x	UI library	Component-based, huge ecosystem
Vite	5.x	Build tool	Faster than CRA, HMR support
React Router DOM	6.x	Routing	Client-side navigation
Axios	1.x	HTTP client	API calls to backend

Language: JavaScript (ES6+) with JSX syntax

1.2 Styling

Tool	Purpose	Why This?
Tailwind CSS	Utility-first CSS	Rapid UI, no custom CSS files
Headless UI	Unstyled components	Accessible dropdowns, modals
Lucide React	Icon library	Modern, lightweight icons
Framer Motion	Animations	Smooth transitions

Language: CSS (via Tailwind classes)

1.3 State Management

Tool	Purpose	When to Use
Redux Toolkit	Global state	Auth, conversations, analytics

Zustand	Lightweight state	Alternative to Redux (simpler)
React Query (TanStack)	Server state	API caching, auto-refetch
Context API	Local state	Theme, socket connection

Recommendation: Redux Toolkit + React Query for production-grade

1.4 Charts & Analytics

Tool	Purpose	Chart Types
Recharts	Data visualization	Bar, Donut, Line, Stacked
Chart.js	Alternative	Simpler API
D3.js	Custom charts	Advanced visualizations

Recommendation: Recharts (matches PRD Module 12)

1.5 Real-Time

Tool	Purpose
Socket.io Client	Live messages, analytics updates
React Socket Context	Centralized socket handling

1.6 Forms & Validation

Tool	Purpose
React Hook Form	Form handling
Zod	Schema validation
Yup	Alternative validation

1.7 UI Components Library

Tool	Purpose
shadcn/ui	Pre-built components (buttons, modals)
Radix UI	Accessible primitives
Ant Design	Full component library (alternative)

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2. BACKEND TECH STACK

2.1 Core Framework

Tool	Version	Purpose
Node.js	20.x LTS	Runtime environment
Express.js	4.x	Web framework
Nodemon	3.x	Dev auto-restart
PM2	5.x	Production process manager

Language: JavaScript (Node.js) — or TypeScript for type safety

2.2 Database

Tool	Purpose	Why This?
MongoDB Atlas	Primary database	Free 512MB, JSON-native
Mongoose	ODM (Object Data Modeling)	Schema validation, relations
Redis (Upstash)	Caching + rate limiting	Free tier, fast
MongoDB Compass	GUI client	Visual DB management

Note: PRD mentions PostgreSQL. Since you're using MERN, MongoDB replaces PostgreSQL — I've mapped all schemas below.

2.3 Authentication & Security

Tool	Purpose
jsonwebtoken (JWT)	Access + refresh tokens
bcryptjs	Password hashing
helmet	HTTP security headers
cors	Cross-origin resource sharing
express-rate-limit	Rate limiting per IP/user
express-validator	Input validation
csurf	CSRF protection
crypto	HMAC-SHA256 signature verification

2.4 Real-Time

Tool	Purpose
Socket.io	WebSocket server
socket.io-redis-adapter	Multi-instance scaling (future)

2.5 AI Integration

Tool	Purpose
@google/generative-ai	Gemini SDK (official)
axios	HTTP client for AI calls
langchain (optional)	Advanced prompt management

Model: Gemini 1.5 Flash (free tier — 15 req/min, 1M tokens/day)

2.6 Message Queue & Jobs

Tool	Purpose
BullMQ	Background job queue
Redis	Queue backend
node-cron	Scheduled tasks (daily reports)

2.7 File Handling

Tool	Purpose
Multer	File upload middleware
Cloudinary	Cloud file storage (free 25GB)
Supabase Storage	Alternative (free 1GB)
sharp	Image optimization

2.8 Email & Notifications

Tool	Purpose
Nodemailer	Email sending (OTP, alerts)
SendGrid	Email API (free 100/day)
Gmail SMTP	Free alternative
Twilio	SMS OTP (optional)

2.9 PDF & CSV Export

Tool	Purpose
PDFKit	PDF generation (server-side)
json2csv	CSV export
exceljs	Excel generation

2.10 Logging & Monitoring

Tool	Purpose
Winston	Structured logging
Morgan	HTTP request logging
Sentry	Error tracking (free 5K/month)
UptimeRobot	Uptime monitoring (free)

2.11 Testing

Tool	Purpose
Jest	Unit testing
Supertest	API testing
Vitest	Fast alternative
Playwright	E2E testing

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3. DATABASE SCHEMA MAPPING (PostgreSQL → MongoDB)

Since PRD was PostgreSQL and you're using MongoDB, here's the mapping:

3.1 Collections (Tables → Collections)

PostgreSQL Table	MongoDB Collection	Mongoose Model
`conversations`	`conversations`	Conversation
`messages`	`messages`	Message
`daily_platform_analytics`	`dailyanalytics`	DailyAnalytics
`hourly_analytics`	`hourlyanalytics`	HourlyAnalytics
`pending_approvals`	`pendingapprovals`	PendingApproval
`leads`	`leads`	Lead
`users`	`users`	User
`audit_logs`	`auditlogs`	AuditLog
`canned_responses`	`cannedresponses`	CannedResponse
`templates`	`templates`	Template
`agent_notes`	`agentnotes`	AgentNote
`business_hours`	`businesshours`	BusinessHours

3.2 Relationships in MongoDB

- PostgreSQL Foreign Keys → MongoDB ObjectId references + Mongoose `populate()`
- Unique Constraints → Mongoose `unique: true` in schema
- Check Constraints → Mongoose custom validators
- Triggers → Mongoose middleware (`pre('save')`, `post('save')`)

Example:

```
// PostgreSQL: conversation_id FOREIGN KEY
// MongoDB:
conversation_id: {
type: mongoose.Schema.Types.ObjectId,
ref: 'Conversation',
required: true
}
```

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4. MODULE-WISE TECH MAPPING

Har PRD module ke liye kaunsi technology use hogi:

PRD Module	Frontend	Backend	Database	External
5. Auth & Onboarding	React + RHF	Express + JWT + bcrypt	`users`	Nodemailer (OTP)
6. Universal Ingestion	—	Express webhooks	`conversations`, `messages`	Telegram/Discord/Meta APIs
7. Interactive Menu	React (preview)	Express + menu service	`messages`	Platform APIs
8. Multi-Lingual AI	—	Express + Gemini SDK	`messages`	Gemini 1.5 Flash
9. Client Dashboard	React + Tailwind + Redux	Express REST APIs	`conversations`, `messages`	Socket.io
10. Lead Extraction	React sidebar	Express regex service	`leads`	—
11. Website Widget	React (embedded)	Express + Socket.io	`conversations`	Socket.io
12. Analytics	Recharts + React Query	Express + Aggregation	`dailyanalytics`, `hourlyanalytics`	—
13. Agent Management	React + Redux	Express RBAC	`users`, `conversations`	—
14. Canned Responses	React + RHF	Express CRUD	`cannedresponses`	—
15. Search & Filter	React + React Query	Express + MongoDB text index	`messages`	—
16. Notes	React	Express CRUD	`agentnotes`	—
17. Business Hours	React form	Express + node-cron	`businesshours`	—
18. WhatsApp Templates	React	Express + Meta API	`templates`	Meta Business API
19. Message Status	React badges	Express webhooks	`messages`	Platform status webhooks
20. Rate Limiting	—	express-rate-limit + Redis	—	Upstash Redis
21. API Integration	Axios	Express routers	—	—
22. Database	—	Mongoose	MongoDB Atlas	—
23. Security	Helmet (frontend headers)	Helmet + JWT + HMAC	—	—
24. Backup	—	mongodump / Atlas backup	MongoDB Atlas	Cloud storage
25. Monitoring	—	Winston + Morgan	—	Sentry + UptimeRobot
26. CI/CD	—	GitHub Actions	—	Render + Vercel
27. Testing	Jest + Playwright	Jest + Supertest	—	—
28. Documentation	—	Swagger / Postman	—	—
31. Channel APIs	—	Express webhooks	—	Telegram, Discord, Meta

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5. LANGUAGE-WISE BREAKDOWN

Language	Where Used	Purpose
JavaScript (ES6+)	Frontend + Backend	Main language (MERN)
JSX	React components	UI markup
TypeScript (optional)	Frontend + Backend	Type safety (recommended)
CSS (Tailwind)	Frontend	Styling
HTML	Frontend	Widget embed code
JSON	API requests/responses	Data format
BSON	MongoDB	Binary JSON storage
Bash / Shell	DevOps	Deployment scripts
YAML	CI/CD	GitHub Actions config
Markdown	Documentation	README, API docs

Note: Python not needed if using MERN. PRD mentioned FastAPI, but we're replacing with Node.js/Express.

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6. AI MODEL & PROMPT STACK

6.1 AI Model

Model	Provider	Free Tier	Use Case
Gemini 1.5 Flash	Google	15 req/min, 1M tokens/day	Primary auto-reply
Gemini 1.5 Pro	Google	Limited	Complex queries (future)
GPT-4o mini	OpenAI	Paid	Alternative
Claude Haiku	Anthropic	Paid	Alternative

Recommendation: Gemini 1.5 Flash (free, fast, multi-lingual)

6.2 Prompt Engineering Layer

Tool	Purpose
System prompts	Business context, rules
Few-shot examples	Language matching
Output parsers	Response trimming (3 sentences)
Guardrails	Anti-hallucination
LangChain (optional)	Prompt chaining

6.3 Language Detection

Tool	Purpose
franc (npm)	Language detection
langdetect	Alternative
Custom regex	Hinglish detection

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7. DEVOPS & INFRASTRUCTURE

7.1 Hosting

Component	Platform	Free Tier
Frontend	Vercel	100GB bandwidth
Backend	Render	750 hrs/month
Backend (alt)	Railway	$5 credit/month
Backend (alt)	Fly.io	3 VMs free
Database	MongoDB Atlas	512MB
Redis	Upstash	10K commands/day
File Storage	Cloudinary	25GB
File Storage (alt)	Supabase Storage	1GB

7.2 CI/CD

Tool	Purpose
GitHub Actions	2000 min/month free
GitHub Container Registry	Docker images
Docker	Containerization
Docker Compose	Local dev

7.3 Monitoring

Tool	Purpose
Sentry	Error tracking
UptimeRobot	Uptime monitoring
Logtail	Log aggregation
Papertrail	Alternative logs

Render Logs	Built-in logs

7.4 Local Development

Tool	Purpose
Ngrok	Expose localhost for webhooks
MongoDB Compass	DB GUI
Postman	API testing
Thunder Client	VS Code API testing
nodemon	Auto-restart

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8. COMPLETE PACKAGE.JSON (Backend)

```
{
"name": "omnihub-backend",
"version": "1.0.0",
"type": "module",
"scripts": {
"dev": "nodemon src/server.js",
"start": "node src/server.js",
"test": "jest",
"lint": "eslint src/"
},
"dependencies": {
"express": "^4.19.2",
"mongoose": "^8.4.0",
"socket.io": "^4.7.5",
"jsonwebtoken": "^9.0.2",
"bcryptjs": "^2.4.3",
"dotenv": "^16.4.5",
"cors": "^2.8.5",
"helmet": "^7.1.0",
"express-rate-limit": "^7.3.0",
"express-validator": "^7.1.0",
"morgan": "^1.10.0",
"winston": "^3.13.0",
"@google/generative-ai": "^0.14.0",
"axios": "^1.7.2",
"bullmq": "^5.8.0",
"ioredis": "^5.4.1",
"multer": "^1.4.5-lts.1",
"cloudinary": "^2.2.0",
"nodemailer": "^6.9.13",
"pdfkit": "^0.15.0",
"json2csv": "^6.0.0-alpha.2",
"exceljs": "^4.4.0",
"node-cron": "^3.0.3",
"@sentry/node": "^8.9.2",
"franc": "^6.2.0",
"helmet": "^7.1.0",
"cookie-parser": "^1.4.6"
},
"devDependencies": {
"nodemon": "^3.1.0",
"jest": "^29.7.0",
"supertest": "^7.0.0",
"eslint": "^9.4.0"
}
}
```

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9. COMPLETE PACKAGE.JSON (Frontend)

```
{
"name": "omnihub-frontend",
"version": "1.0.0",
"type": "module",
"scripts": {
"dev": "vite",
"build": "vite build",
"preview": "vite preview",
"lint": "eslint src/"
},
"dependencies": {
"react": "^18.3.1",
"react-dom": "^18.3.1",
"react-router-dom": "^6.23.1",
"axios": "^1.7.2",
"redux": "^5.0.1",
"react-redux": "^9.1.2",
"@reduxjs/toolkit": "^2.2.5",
"@tanstack/react-query": "^5.40.0",
"socket.io-client": "^4.7.5",
"recharts": "^2.12.7",
"react-hook-form": "^7.51.5",
"zod": "^3.23.8",
"@hookform/resolvers": "^3.6.0",
"tailwindcss": "^3.4.4",
"lucide-react": "^0.395.0",
"framer-motion": "^11.2.10",
"@headlessui/react": "^2.1.1",
"clsx": "^2.1.1",
"date-fns": "^3.6.0",
"react-hot-toast": "^2.4.1",
"jspdf": "^2.5.1",
"papaparse": "^5.4.1"
},
"devDependencies": {
"vite": "^5.2.13",
"@vitejs/plugin-react": "^4.3.1",
"eslint": "^9.4.0",
"autoprefixer": "^10.4.19",
"postcss": "^8.4.38"
}
}
```

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10. FOLDER STRUCTURE

10.1 Backend

```
omnihub-backend/
├── src/
│   ├── server.js
│   ├── app.js
│   ├── config/
│   │   ├── db.js
│   │   ├── redis.js
│   │   └── cloudinary.js
│   ├── models/
│   │   ├── Conversation.js
│   │   ├── Message.js
│   │   ├── DailyAnalytics.js
│   │   ├── HourlyAnalytics.js
```

```
│   │   ├── PendingApproval.js
│   │   ├── Lead.js
│   │   ├── User.js
│   │   ├── AuditLog.js
│   │   ├── CannedResponse.js
│   │   ├── Template.js
│   │   ├── AgentNote.js
│   │   └── BusinessHours.js
│   ├── routes/
│   │   ├── webhooks/
│   │   │   ├── telegram.js
│   │   │   ├── discord.js
│   │   │   ├── whatsapp.js
│   │   │   ├── instagram.js
│   │   │   └── linkedin.js
│   │   ├── dashboard/
│   │   │   ├── conversations.js
│   │   │   ├── messages.js
│   │   │   ├── approvals.js
│   │   │   ├── analytics.js
│   │   │   ├── leads.js
│   │   │   └── agents.js
│   │   ├── ai.js
│   │   └── auth.js
│   ├── controllers/
│   ├── services/
│   │   ├── normalizer.js
│   │   ├── aiService.js
│   │   ├── menuService.js
│   │   ├── mediaService.js
│   │   ├── analyticsService.js
│   │   ├── leadExtractor.js
│   │   ├── dispatcher.js
│   │   └── emailService.js
│   ├── middleware/
│   │   ├── auth.js
│   │   ├── rateLimiter.js
│   │   ├── validateWebhook.js
│   │   └── errorHandler.js
│   ├── sockets/
│   │   └── index.js
│   ├── jobs/
│   │   ├── dailyReport.js
│   │   └── cleanup.js
│   └── utils/
│       ├── logger.js
│       ├── security.js
│       └── helpers.js
├── tests/
├── .env
├── package.json
└── README.md
```

10.2 Frontend

```
omnihub-frontend/
├── public/
│   ├── favicon.ico
│   ├── logo.svg
│   └── widget.js              ← Embeddable chat widget (Module 11)
│
├── src/
```

```
│   ├── main.jsx                ← Entry point
│   ├── App.jsx                 ← Root component
│   │
│   ├── api/                    ← API Layer
│   │   ├── axios.js            ← Axios instance + interceptors
│   │   ├── authApi.js          ← Auth endpoints
│   │   ├── conversationApi.js  ← Conversation endpoints
│   │   ├── analyticsApi.js     ← Analytics endpoints
│   │   ├── approvalApi.js      ← Approval endpoints
│   │   ├── leadApi.js          ← Lead endpoints
│   │   └── aiApi.js            ← AI endpoints
│   │
│   ├── store/                  ← Redux Toolkit
│   │   ├── index.js            ← Store config
│   │   ├── authSlice.js        ← Module 5: Auth
│   │   ├── conversationSlice.js ← Module 9: Conversations
│   │   ├── messageSlice.js     ← Module 9: Messages
│   │   ├── approvalSlice.js    ← Module 9: Approvals
│   │   ├── analyticsSlice.js   ← Module 12: Analytics
│   │   ├── leadSlice.js        ← Module 10: Leads
│   │   ├── agentSlice.js       ← Module 13: Agents
│   │   ├── cannedResponseSlice.js ← Module 14: Canned
│   │   ├── noteSlice.js        ← Module 16: Notes
│   │   ├── businessHoursSlice.js ← Module 17: Hours
│   │   ├── templateSlice.js    ← Module 18: Templates
│   │   └── sessionSlice.js     ← Module 5.3: Sessions
│   │
│   ├── context/                ← React Context
│   │   ├── SocketContext.jsx   ← Socket.io provider
│   │   ├── ThemeContext.jsx    ← Dark/light mode
│   │   └── NotificationContext.jsx ← Toast/notifications
│   │
│   ├── routes/                 ← Route guards + config
│   │   ├── AppRoutes.jsx       ← All routes
│   │   ├── ProtectedRoute.jsx  ← Auth guard
│   │   ├── PublicRoute.jsx     ← Redirect if logged in
│   │   └── RoleRoute.jsx       ← Role-based guard (Module 5.4)
│   │
│   ├── pages/                  ← Page components
│   │   ├── auth/               ← Module 5: Authentication
│   │   │   ├── Register.jsx
```

```
│   │   ├── VerifyEmail.jsx
│   │   ├── Login.jsx
│   │   ├── VerifyOTP.jsx
│   │   ├── BackupCodes.jsx
│   │   ├── ForgotPassword.jsx
│   │   ├── ResetPassword.jsx
│   │   └── SessionsManager.jsx
│   │
│   ├── dashboard/            ← Module 9: Dashboard
│   │   ├── Dashboard.jsx     ← Main 3-panel view
│   │   ├── ConversationView.jsx
│   │   └── Approvals.jsx
│   │
│   ├── analytics/            ← Module 12: Analytics
│   │   ├── Analytics.jsx
│   │   └── Reports.jsx
│   │
│   ├── leads/                ← Module 10: Leads
│   │   └── LeadsList.jsx
│   │
│   ├── agents/               ← Module 13: Agents
│   │   ├── AgentsList.jsx
│   │   └── AgentDetail.jsx
│   │
│   ├── settings/             ← Settings pages
│   │   ├── Profile.jsx
│   │   ├── BusinessHours.jsx    ← Module 17
│   │   ├── CannedResponses.jsx  ← Module 14
│   │   ├── Templates.jsx        ← Module 18
│   │   ├── Channels.jsx         ← Module 31: Channel connect
│   │   ├── Sessions.jsx         ← Module 5.3
│   │   └── Security.jsx         ← Module 5.2: 2FA
│   │
│   ├── onboarding/           ← Module 30: Client setup
│   │   ├── OnboardingWizard.jsx
│   │   ├── KnowledgeBase.jsx
│   │   └── ChannelConnect.jsx
│   │
│   ├── NotFound.jsx
│   └── ServerError.jsx
│
```

```
│   ├── components/              ← Reusable components
│   │   ├── common/              ← Shared UI
│   │   │   ├── Button.jsx
│   │   │   ├── Input.jsx
│   │   │   ├── Modal.jsx
│   │   │   ├── Dropdown.jsx
│   │   │   ├── Badge.jsx
│   │   │   ├── Avatar.jsx
│   │   │   ├── Spinner.jsx
│   │   │   ├── Tooltip.jsx
│   │   │   ├── Pagination.jsx
│   │   │   ├── EmptyState.jsx
│   │   │   ├── ErrorBoundary.jsx
│   │   │   └── ConfirmDialog.jsx
│   │   │
│   │   ├── layout/              ← Layout components
│   │   │   ├── AuthLayout.jsx       ← Module 5: Auth pages wrapper
│   │   │   ├── DashboardLayout.jsx  ← Main dashboard wrapper
│   │   │   ├── Sidebar.jsx
│   │   │   ├── Topbar.jsx
│   │   │   └── Footer.jsx
│   │   │
│   │   ├── auth/                ← Module 5: Auth components
│   │   │   ├── PasswordInput.jsx
│   │   │   ├── PasswordStrength.jsx
│   │   │   ├── OTPInput.jsx
│   │   │   ├── ResendTimer.jsx
│   │   │   ├── BackupCodeCard.jsx
│   │   │   ├── SessionCard.jsx
│   │   │   ├── TwoFASetupCard.jsx
│   │   │   └── LoginAuditRow.jsx
│   │   │
│   │   ├── dashboard/           ← Module 9: Dashboard
│   │   │   ├── LeftPanel.jsx        ← Chat list
│   │   │   ├── CenterPanel.jsx      ← Conversation view
│   │   │   ├── RightPanel.jsx       ← Lead details
│   │   │   ├── ChatBubble.jsx
│   │   │   ├── MessageInput.jsx
│   │   │   ├── AutoPilotToggle.jsx
│   │   │   ├── ApprovalCard.jsx
│   │   │   ├── AI PolishButton.jsx
```

```
│   │   │   └── ChannelBadge.jsx
│   │   │
│   │   ├── analytics/          ← Module 12: Analytics
│   │   │   ├── KpiCard.jsx
│   │   │   ├── BarChart.jsx        ← Inbound vs Outbound
│   │   │   ├── DonutChart.jsx      ← Channel distribution
│   │   │   ├── LineChart.jsx       ← Hourly traffic
│   │   │   ├── StackedChart.jsx    ← Response breakdown
│   │   │   ├── DateRangePicker.jsx
│   │   │   ├── PlatformTable.jsx
│   │   │   └── ExportButton.jsx
│   │   │
│   │   ├── leads/              ← Module 10: Leads
│   │   │   ├── LeadCard.jsx
│   │   │   ├── LeadTags.jsx
│   │   │   └── LeadStatusBadge.jsx
│   │   │
│   │   ├── agents/             ← Module 13: Agents
│   │   │   ├── AgentCard.jsx
│   │   │   ├── AgentStatusBadge.jsx
│   │   │   └── AgentPerformance.jsx
│   │   │
│   │   ├── settings/           ← Settings components
│   │   │   ├── BusinessHoursForm.jsx
│   │   │   ├── CannedResponseForm.jsx
│   │   │   ├── TemplateForm.jsx
│   │   │   └── ChannelConnectCard.jsx
│   │   │
│   │   ├── widget/             ← Module 11: Website widget
│   │   │   ├── ChatWidget.jsx
│   │   │   ├── PreChatForm.jsx
│   │   │   ├── TypingIndicator.jsx
│   │   │   └── WidgetLauncher.jsx
│   │   │
│   │   └── notes/              ← Module 16: Notes
│   │       ├── NotesList.jsx
│   │       ├── NoteCard.jsx
│   │       └── MentionInput.jsx
│   │
│   ├── hooks/                  ← Custom hooks
│   │   ├── useAuth.js          ← Module 5
```

```
│   │   ├── useOTPTimer.js          ← Module 5.2
│   │   ├── useIdleTimer.js         ← Module 5.3
│   │   ├── usePasswordStrength.js  ← Module 5.1
│   │   ├── useSessionRefresh.js    ← Module 5.3
│   │   ├── useSocket.js            ← Socket connection
│   │   ├── useAnalytics.js         ← Module 12
│   │   ├── useConversations.js     ← Module 9
│   │   ├── useApprovals.js         ← Module 9.2
│   │   ├── useDebounce.js
│   │   ├── useLocalStorage.js
│   │   ├── useMediaQuery.js
│   │   └── useClickOutside.js
│   │
│   ├── services/                   ← API service layer
│   │   ├── authService.js          ← Module 5
│   │   ├── conversationService.js  ← Module 9
│   │   ├── analyticsService.js     ← Module 12
│   │   ├── approvalService.js      ← Module 9.2
│   │   ├── leadService.js          ← Module 10
│   │   ├── aiService.js            ← Module 8
│   │   ├── agentService.js         ← Module 13
│   │   ├── cannedResponseService.js ← Module 14
│   │   ├── noteService.js          ← Module 16
│   │   ├── businessHoursService.js ← Module 17
│   │   ├── templateService.js      ← Module 18
│   │   ├── sessionService.js       ← Module 5.3
│   │   └── channelService.js       ← Module 31
│   │
│   ├── utils/                      ← Utilities
│   │   ├── validators.js           ← Email, password regex
│   │   ├── tokenStorage.js         ← JWT storage
│   │   ├── deviceInfo.js           ← Browser/device detection
│   │   ├── formatters.js           ← Date, number formats
│   │   ├── constants.js            ← App constants
│   │   ├── permissions.js          ← Role-based permissions
│   │   ├── chartColors.js          ← Analytics colors
│   │   ├── exportUtils.js          ← PDF/CSV export
│   │   └── socketEvents.js         ← Socket event names
│   │
│   └── assets/                     ← Static assets
│       ├── images/
```

```
│   ├── icons/
│   └── fonts/
│
├── .env                    ← Environment variables
├── .env.example
├── .eslintrc.js
├── .prettierrc
├── .gitignore
├── index.html
├── tailwind.config.js
├── postcss.config.js
├── vite.config.js
├── package.json
└── README.md---
```

11. ENVIRONMENT VARIABLES

Backend `.env`

```
# Server
PORT=5000
NODE_ENV=development

# Database
MONGODB_URI=mongodb+srv://...
REDIS_URL=redis://...

# JWT
JWT_SECRET=your_secret_key
JWT_REFRESH_SECRET=your_refresh_secret
JWT_EXPIRES=15m
JWT_REFRESH_EXPIRES=7d

# AI
GEMINI_API_KEY=your_gemini_key

# Platforms
TELEGRAM_BOT_TOKEN=...
DISCORD_BOT_TOKEN=...
META_APP_SECRET=...
META_VERIFY_TOKEN=...
WHATSAPP_PHONE_ID=...
INSTAGRAM_ACCOUNT_ID=...

# Storage
CLOUDINARY_CLOUD_NAME=...
CLOUDINARY_API_KEY=...
CLOUDINARY_API_SECRET=...

# Email
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=...
SMTP_PASS=...

# Monitoring
```

```
SENTRY_DSN=...

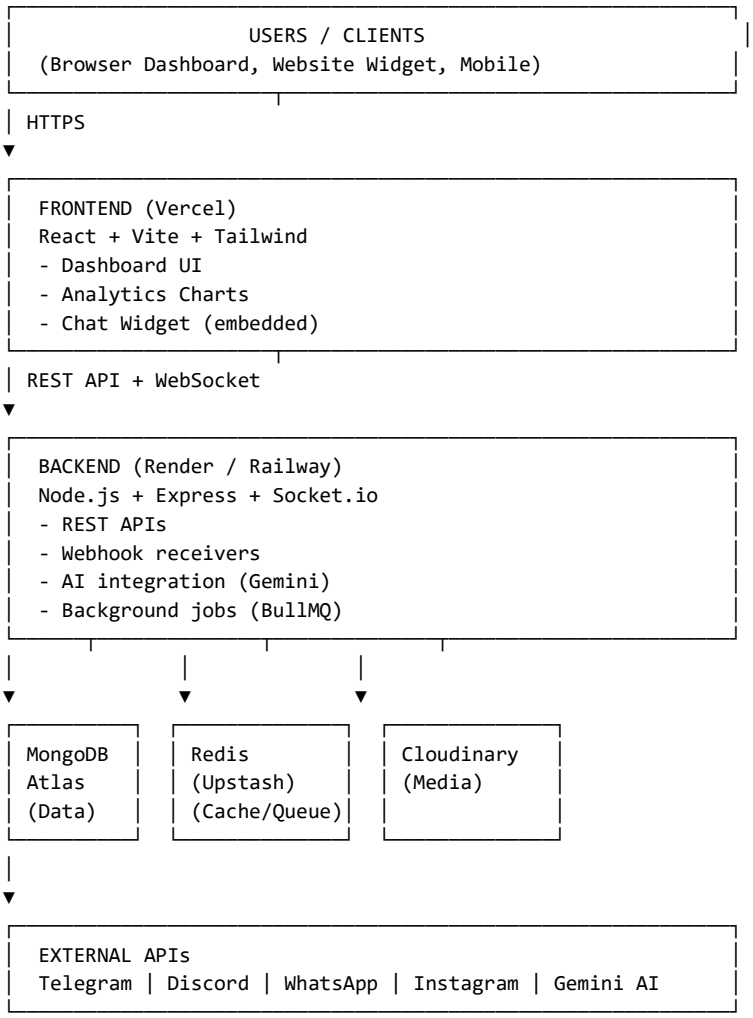
# Frontend URL (for CORS)
CLIENT_URL=http://localhost:5173
```

Frontend `.env`

```
VITE_API_URL=http://localhost:5000/api
VITE_SOCKET_URL=http://localhost:5000
VITE_GEMINI_KEY=...
```

---

12. DEPLOYMENT ARCHITECTURE



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13. COST BREAKDOWN (All Free Tier)

Service	Free Tier	Cost
Vercel (Frontend)	100GB bandwidth	₹0
Render (Backend)	750 hrs/month	₹0
MongoDB Atlas	512MB	₹0
Upstash Redis	10K commands/day	₹0
Cloudinary	25GB storage	₹0
Gemini API	15 req/min	₹0
GitHub Actions	2000 min/month	₹0
Sentry	5K events/month	₹0

UptimeRobot	50 monitors	₹0
Ngrok	1 tunnel	₹0
TOTAL		₹0

---

14. RECOMMENDED VS CODE EXTENSIONS

- ESLint
- Prettier
- Tailwind CSS IntelliSense
- MongoDB for VS Code
- Thunder Client
- GitLens
- Error Lens
- Auto Rename Tag
- ES7+ React Snippets

---

15. FINAL RECOMMENDATION SUMMARY

Layer	Choice	Reason
Frontend Framework	React 18 + Vite	Fast, modern
Styling	Tailwind CSS	Rapid UI
State	Redux Toolkit + React Query	Best combo
Charts	Recharts	Matches PRD
Backend	Node.js + Express	MERN standard
Database	MongoDB Atlas	Free, flexible
ODM	Mongoose	Best MongoDB ORM
Real-time	Socket.io	Industry standard
AI	Gemini 1.5 Flash	Free, fast
Auth	JWT + bcrypt	Standard
Cache/Queue	Redis (Upstash) + BullMQ	Free tier
Files	Cloudinary	Free 25GB
Email	Nodemailer + Gmail	Free
Hosting	Vercel + Render	Both free
